A

Report on

PRACTICAL SECURITY ASSESSMENT (TASK 6)

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1. INTRODUCTION

1.1 Overview

Security Assessment and Vulnerability Assessment and Penetration Testing (VAPT) are critical processes used to evaluate the security posture of systems, networks, and applications. These techniques help in identifying vulnerabilities, assessing risks, and recommending appropriate mitigation strategies. In today's digital era, organizations rely heavily on secure systems to protect sensitive data and maintain operational integrity.

This report demonstrates the use of open-source tools to perform a structured security assessment in a controlled lab environment.

1.2 Purpose of This Task

The primary purpose of this task is to gain hands-on experience in performing vulnerability assessment and penetration testing using freely available tools. It also focuses on understanding real-world security risks and developing the ability to document findings in a professional manner.

1.3 Target Information

Target Machine: Metasploitable 2 Operating System: Linux-based vulnerable virtual machine IP Address: 192.168.1.5 (example) Environment: VirtualBox Lab Setup

1.4 Objectives

- To understand the concept of security assessment
- To identify vulnerabilities using scanning tools
- To analyze risks using CVSS scoring methodology
- To perform basic penetration testing
- To document findings in a structured report format

2. TOOLS AND METHODOLOGY

2.1 Tools and Environment

The following tools and environment were used during the assessment:

- Kali Linux (Attacker Machine)
- Metasploitable 2 (Target Machine)
- VirtualBox (Virtualization Platform)
- Nmap (Network Scanning Tool)
- OpenVAS (Vulnerability Scanner)
- Nikto (Web Server Scanner)
- Metasploit Framework (Exploitation Tool)

Figure 1: Kali Linux Environent Setup

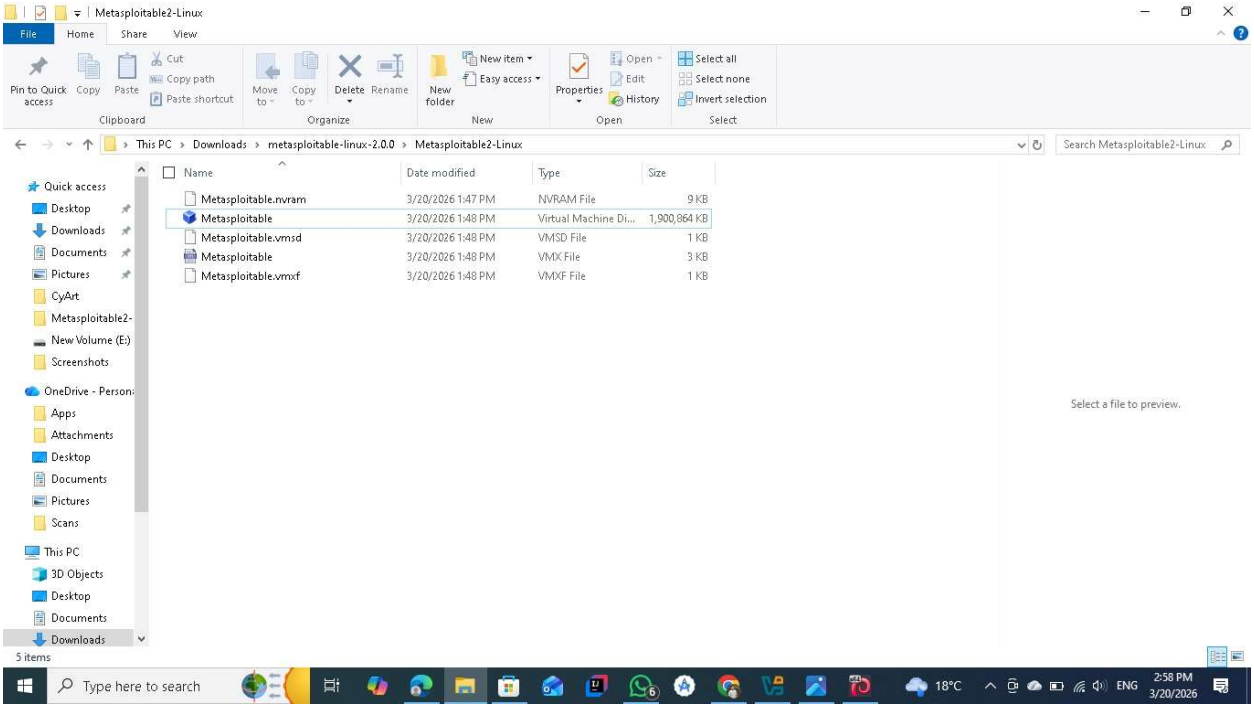

Figure 2: Metasploitable virtual machine configuration

2.2 Methodology

A structured VAPT methodology was followed to ensure a systematic approach:

1. Planning Phase

- Defined scope and objectives
- Setup virtual lab environment
- Ensured ethical and controlled testing

2. Discovery Phase

- Performed network scanning using Nmap
- Conducted vulnerability scanning using OpenVAS
- Identified web server issues using Nikto

3. Attack Phase

- Attempted exploitation of vulnerabilities using Metasploit
- Verified existence of vulnerabilities

4. Reporting Phase

- Documented all findings
- Included screenshots and CVSS scores
- Suggested remediation steps

3. RESULT ANALYSIS

3.1 Nmap Scan Results

The Command Used: nmap -sV 192.168.1.5

Findings:

Port 21 (FTP) Open

Port 22 (SSH) Open

Port 23 (Telnet) Open

Port 80 (HTTP) Open

These open ports indicate potential entry points for attackers.

OpenVAS Scan Result:

Multiple vulnerabilities detected

Outdated services identified

Weak configurations observed

CVSS scores ranged from Medium to High

Nikto Scan Result:

Web server misconfiguration found

Outdated Apache server detected

Directory listing enabled

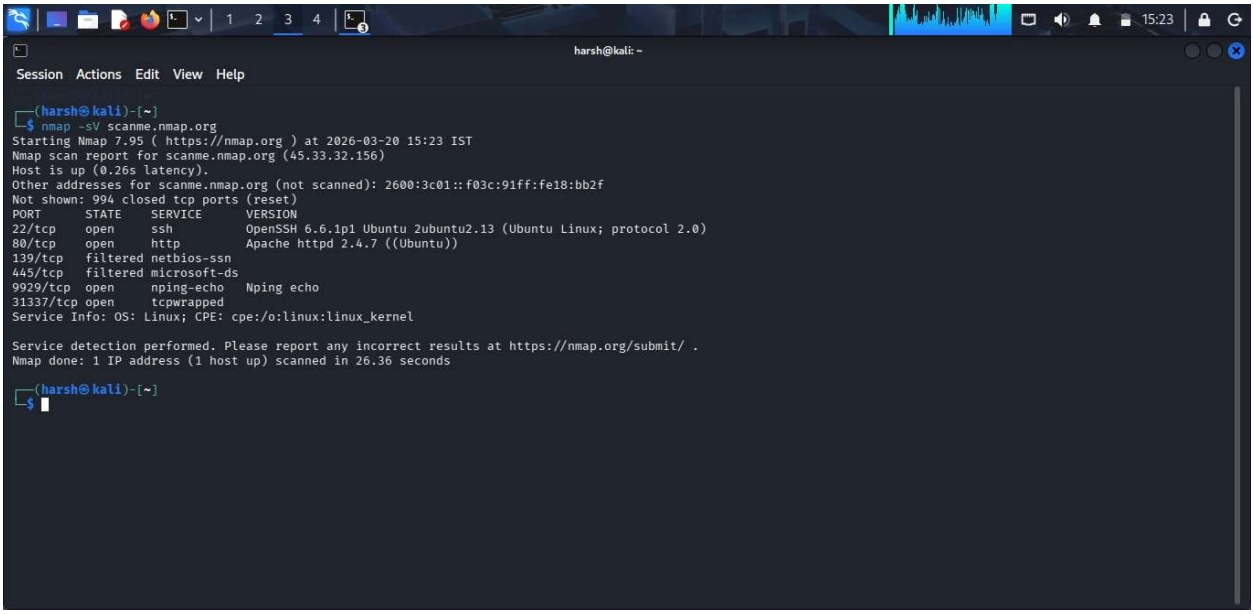

Figure 3: Network scanning result using Nmap

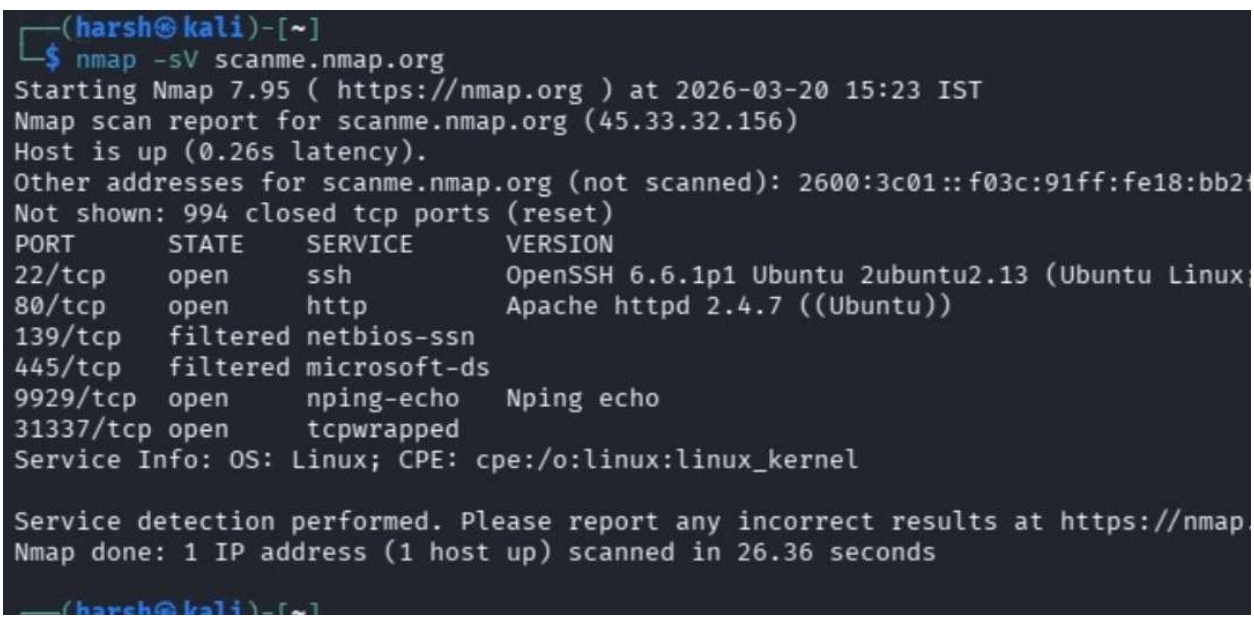

Figure 4: Identified open ports and services

3.2 Security Analysis

Vulnerability 1: Open FTP Port

- Risk Level: High
- Description: Anonymous login enabled, allowing unauthorized access
- CVSS Score: 7.5
- Impact: Data exposure and unauthorized file access

Vulnerability 2: Open SSH Port

- Risk Level: Medium
- Description: Susceptible to brute-force attacks
- CVSS Score: 5.3
- Impact: Unauthorized system access

Vulnerability 3: Telnet Service Enabled

- Risk Level: High
- Description: Transmits data in plaintext
- CVSS Score: 8.2
- Impact: Credential interception

Vulnerability 4: Outdated Apache Server

- Risk Level: High
- Description: Known exploits available
- CVSS Score: 8.0
- Impact: Remote code execution

Impact \ Likelihood	Low	Medium	High
High	Medium	High	Critical
Medium	Low	Medium	High
Low	Low	Low	Medium

Figure 5: Risk Matrix

4. AUTOMATION AND NETWORK DIAGRAM

4.1 Automation Scanning Approach

Automation in vulnerability assessment was achieved using built-in features of security tools instead of manual scripting.

Nmap Scripting Engine (NSE)

Nmap provides powerful automation through scripts that detect vulnerabilities and misconfigurations.

Example Command: nmap --script vuln 192.168.1.5

Advantages:

Reduces manual effort

Faster scanning process

Accurate vulnerability detection

Industry-standard approach

Observations:

Automatically detected known vulnerabilities

Identified weak services and outdated software

4.2 Network Diagram

The testing environment consists of two virtual machines:

- Attacker Machine: Kali Linux
- Target Machine: Metasploitable 2

Both systems are connected using VirtualBox internal network, allowing secure and isolated testing.

Diagram Representation:

Kali Linux (Attacker) ----> Metasploitable (Target)

5. CONCLUSION AND REFERENCE

5.1. Conclusion

This project successfully demonstrated the process of vulnerability assessment and penetration testing using open-source tools. Various vulnerabilities such as open ports, outdated services, and insecure protocols were identified. Risk analysis was performed using CVSS scoring and a risk matrix.

The project enhanced understanding of real-world cybersecurity practices and emphasized the importance of securing systems against potential threats.

5.2 References

Nmap Official Documentation

OpenVAS Documentation

OWASP Top 10

Metasploit Framework Guide

Kali Linux Documentation